



Feasibility study of a standalone hybrid energy system to supply electricity to a rural community in South Sudan

Nyagong Santino David Ladu^{a,*}, Ravi Samikannu^b,
Kahsu Gebrehans Gebreslassie^c, Martin Sankoh^d, Lemi Edward Rufas Hakim^f,
Alsadig Badawi^a, Talib Paskwali Beshir Latio^e

^a Department of Mathematics and Physics, Rumbek University of Science and Technology, Rumbek, South Sudan

^b Department of Electrical, Computer and Telecommunication Engineering, Botswana International University of Science & Technology, Palapye, Botswana

^c Department of Mechanical Engineering, King Fahd University of Petroleum and Minerals, Saudi Arabia

^d Department of Mechanical and Maintenance Engineering, Fourah Bay College - University of Sierra Leone, Sierra Leone

^e Department of Energy Engineering, Pan African University Institute of Water and Energy Sciences - PAUWES, c/o Tlemcen University, B.P. 119 | Pôle Chetouane, Tlemcen 13000, Algeria

^f Department of Biology and Chemistry, Rumbek University of Science and Technology, Rumbek, South Sudan

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ABSTRACT

Despite promising solar potential in South Sudan, rural electrification has long been an issue for the country's growth and development, as well as addressing climate change and fuel cost limits. This study aims at the feasibility analysis of a hybrid energy system for a rural community in the Southern part of South Sudan without access to electricity. Over a year, typical energy consumption profiles were generated based on the energy needs of the community. HOMER pro program was used to configure and optimize the system, and six different combinations were simulated and analyzed economically and technically based on the standalone mode of operation. The PV/DG/Battery design offers the lowest Net Present Cost (NPC) and Cost of Energy (COE), with a 22.94% return on investment due to the substantial solar potential. The study also found a modest wind speed in rural areas. In the rural community, the proposed configuration system can help provide electricity access with an 85% renewable fraction. The results of this study's six hybrid systems/combinations can serve as a reference benchmark for stakeholders, demonstrating the possibilities of boosting renewables, changing configuration settings, and lowering the cost of electricity.

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* Corresponding author.

E-mail addresses: nyagongsantino19@gmail.com (N.S.D. Ladu), ravis@biust.ac.bw (R. Samikannu), kas0914584389@gmail.com (K.G. Gebreslassie), sankohmartinsidi@gmail.com (M. Sankoh), lemiedward@yahoo.com (L.E.R. Hakim), alsadiga352@gmail.com (A. Badawi), talibpaskwali03@gmail.com (T.P.B. Latio).

Introduction

The use of renewable energy sources in electricity generation has sparked widespread attention due to advantages such as its favorable environmental effects, affordability and reliability, and capacity to be renewed and reused. The inclusion of a generator in isolated hybrid systems is to improve the continuous flow of electricity because solar and wind are intermittent sources of energy, with the sun resource available only during the day and the majority of the wind resource mostly available at night, and the stored energy in the battery system may not be enough to meet the load. South Sudan lies on the coordinates 6°52.6'N and 31°18.4'E. In 2019, conventional sources such as diesel generators represent more than 99% of electricity generation in South Sudan with a capacity estimated at 204 MW, whereas solar accounts for only an estimated 1 MW of capacity, which accounts for less than 1% of electricity generation in the country [1]. In South Sudan's rural communities, kerosene lamps, firewood, crop wastes, charcoal, and animal dung are the most frequent sources of energy for lighting, heating, and cooking. Geographical constraints, insufficient markets, political barriers, and other issues may hamper the development of power generation projects or grid electricity expansion in rural places.

Numerous studies on hybrid energy systems have been conducted using the HOMER tool for various remote locations in Africa. The majority of earlier studies on rural hybrid energy systems were primarily focused on technical, economic, and feasibility studies. A feasibility analysis of a stand-alone PV/wind/generator hybrid system for a rural location in Comoros to identify the most optimal solution revealed that combining wind and diesel is the most viable and cost-effective alternative [2]. Meanwhile, a study presented by Bentouba and Bourouis [3] on the feasibility of a wind-PV hybrid power generation system for a rural area in the South of Algeria that is not connected to the utility grid shows that a diesel generator is required as a backup system to meet 100% of the rural area's electricity demand, as well as a reduction in CO₂ emissions of about 593.125 tons per year. A feasibility study of a stand-alone PV/DG/battery hybrid energy system for isolated areas in northern Ghana revealed a system that is optimized, cost-effective, and environmentally benign [4].

The fact that renewable energy accounts for barely 1% of power generation in South Sudan highlights the necessity of this study in aligning with the government of South Sudan's targets of increasing electricity access while considering renewable energy sources.

Until recently, only a few small standalone solar photovoltaic installations have been installed in South Sudan, mostly in urban areas to power radio stations and water pumps. One of the major projects accomplished in 2020 was the installation of a hybrid solar PV-diesel system at the United Nations House compound in Juba, the capital city of South Sudan [5]. As a result, the main goal of this research is to explore the possibility of using a hybrid power system to deliver electricity to a rural area in South Sudan while simultaneously cutting fuel costs and emissions.

This is the first research of its sort in the domain of hybrid energy systems for a typical South Sudan rural area to be presented as such the study also intends to address a research vacuum in electricity accessibility through hybrid energy systems in typical rural areas of South Sudan's Southern rural areas, as well as serve as a roadmap for hybrid energy systems implementation in such areas. The study will investigate the technical and economic parameters of several standalone hybrid energy system configurations to determine the most cost-effective and reliable standalone hybrid energy system for addressing socio-economic development challenges through affordable and reliable electricity. By taking into consideration the potential of the site resources, the hybrid energy systems will be optimized both economically and technically.

Location description

A brief geographic description of the Payam

Nyepo Payam (Payams in South Sudan are the second-lowest administrative levels after counties) is found in the northern part of Kajo-keji country of Central Equatoria State, Equatoria region, South Sudan. The Payam is made up of four bomas (Bomas are the smallest administrative units under Payams) including Kansuk Boma, Rodo Boma, Lori Boma, and Gederu Boma. Kajo-keji county is located in the Southern part of South Sudan bordered by Juba County to the north, Lainya County to the west, Magwi County (Eastern Equatoria State) to the east, and Uganda to the South. Fig. 1 shows the location of Nyepo Payam in Kajo-Keji county on a map of South Sudan.

Load demand estimation of the Payam

According to a survey report in Kajo-Keji County in 2019 by South Sudan - Water, Sanitation and Hygiene (WASH) - a non-profit organization, Nyepo Payam has an estimated population of 2010 individuals and 335 households. There is currently no installed power system that provides electricity to the Payam.

The estimated load is considered on a 24 hours load consumption from the following categories: households, schools, health centers, churches, and retail shops, with appliances such as ceiling fans, lamps (light bulbs), televisions (TV), mobile phones, radios, fridges, laptops, desktop computers, speakers, printers, and microscopes making up the load consumption. The average daily power load is estimated to be 778.25 kWh/day, with a daily peak of 180.46 kW and a load factor of 17%. Peak load is encountered in the afternoon hours of 13:00, 14:00, and 15:00 when the ceiling fans are turned on, per the daily load profile. The load was modeled as an AC load.

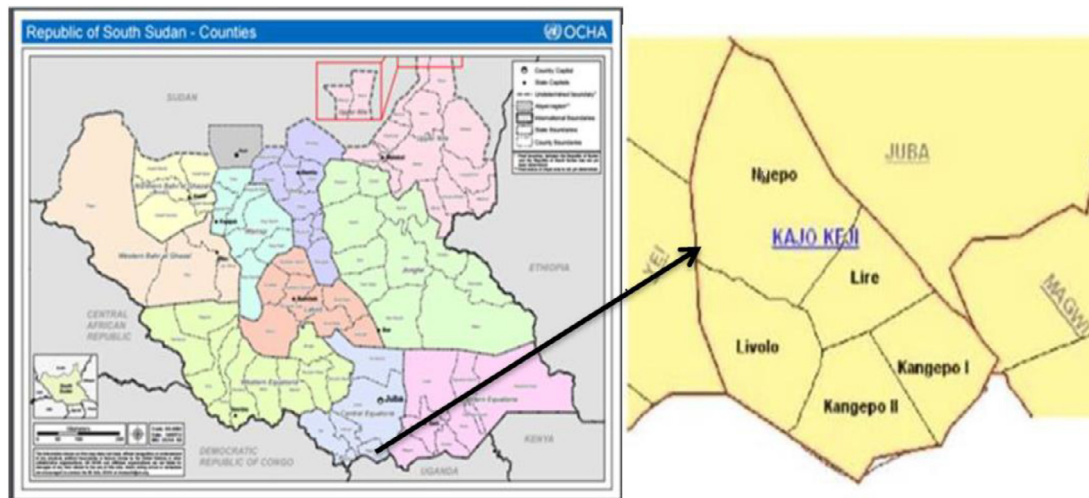


Fig. 1. Map of South Sudan (left enclosed in square) showing Kajo-Keji County and its Payams including Nyepo Payam in the north (right) [6].

Table 1

Average daily solar radiation data.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Clearness Index	6.19	6.39	5.99	5.66	5.51	4.97	4.69	4.95	5.56	5.4	5.47	5.81
Daily Radiation (kW/m ² /day)	0.642	0.632	0.574	0.546	0.552	0.512	0.479	0.488	0.538	0.532	0.564	0.615

Table 2

Average daily wind speed data.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Average wind speed (m/s)	4.94	4.99	4.56	3.84	3.03	2.7	2.47	2.6	2.8	2.9	3.47	4.44

Domestic electrical appliance possession varies depending on the household's income level. Because seasonal variation consumption effects are only felt on a few days during a few specific months during the rainy season, load consumption is considered constant throughout the year. As a result, these changes were ignored during the load modeling of this study.

The solar energy potential of the Payam

The climate in South Sudan is tropical and humid, with temperatures regularly topping 25 °C. The country has an abundant solar potential that is suitable for both small-scale and large-scale solar installations. Kajo-keji county is located between 3°51.3'N and 31°39.4'E.

The clearness index and solar radiation for the site indicated in Table 1 were obtained from NASA's worldwide horizontal irradiation resource data. The yearly average solar irradiation at the area is 5.55 kW/m²/day, with the highest average daily radiation of 6.39 kW/m²/day occurring in February and the lowest average daily radiation of 4.69 kW/m²/day occurring in July.

Wind potential of the Payam

As per a preliminary study on wind resource assessment conducted in the country, South Sudan's Southern region, including Kajo-keji county, and some northern parts, has the country's lowest wind potential [7].

The monthly average wind speed data were likewise collected from the NASA resource website is presented in Table 2. The yearly average wind speed, measured from a height of 50 m above the ground, is 3.56 m/s. The maximum monthly average daily wind speed is 4.99 m/s in February, while the lowest monthly average daily wind speed is 2.47 m/s in July.

Temperature data of the Payam

Temperature data is used in the modeling of PV modules and wind since temperature influences their operating efficiency. Temperature data was also acquired from the NASA data resource website. The scaled annual average temperature is 26.23 °C. Table 3 presents the average monthly temperature of the site.

Table 3
Average monthly temperature data.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature	28.24	29.54	29.88	27.9	25.63	24.43	23.69	23.65	24.3	24.91	25.65	26.88

Table 4
Fuel properties.

Fuel: Diesel	
Fuel curve intercept	0.825 L/h
Fuel curve slope	0.273 L/h/kW
Emissions	
CO (g/L fuel)	16.34
Unburned HC (g/L fuel)	0.72
Particulates (g/L fuel)	0.098
Fuel Sulfur to PM (%)	2.2
NO _x (g/L fuel)	15.359
Fuel Properties	
Lower Heating Value (MJ/kg)	43.2
Density (kg/m ³)	820
Carbon Content (%)	88
Sulfur Content (%)	0.4

The months of January, February, March, and April have higher temperatures than the rest of the year. The lowest temperature, 23.69 °C, is recorded in July.

Methodology and system configuration

This section encompasses the system setup components.

Excel was used to model the primary data from the location. The calculated total power of each appliance for the various load consumption categories is then added together to obtain the total power demand for each hour of the day. The total power demand for each hour of the day is then entered into the HOMER pro program for each of the 12 months of the year. These inputted data simulate the scaled average load demand, a scaled peak load, and load factor.

System configuration

This study uses HOMER PRO software for systems simulation and optimization. By comparing the power demand in each iteration level to the electricity that the system can generate in a given time interval, HOMER models can undertake a techno-economic feasibility analysis over a year [8]. The components in this study including the PV system, the wind system, the generator setup, the battery system, and the converter were modeled and configured using the HOMER Pro-software.

Photovoltaic (PV) system

For this investigation, a rated 315 W polycrystalline TrinTall flat-plate PV modules with 72 cells and a 16.2% efficiency are used. The expenses of capital and replacement are set at \$400 per kW and \$350 per kW, respectively. The annual operating and maintenance (O&M) cost is set at \$10 per kW. The operational temperature of the cell is 44 °C, and its temperature coefficient is -0.410.

PV cells are connected in series and parallel to provide the desired current and voltage.

Wind turbine

A 3 bladed, 1.5 kW rated capacity wind turbine was selected and manufactured by the Australian Wind and Solar (AWS) manufacturing company. The capital and replacement expenditures are respectively \$6000 and \$5300 per kW. \$20 per year was set as the O&M. The lifetime of this turbine is 20 years and its hub height is 12 m. Also, from the datasheet, the peak power of this turbine is 1.7 kW, a rated speed of 10.5 m/s, and a tip ratio of 8.5.

Diesel generator (DG)

For this simulation, a 25 kW fixed capacity generic Genset generator is used, with a capital cost of around \$8000 and an O&M cost of \$0.010 per hour. For a 15,000 h operating life, the minimum load ratio taken into account is 25%. Table 4 shows the properties of the fuel utilized in the simulation.

The HOMER simulated fuel consumption and efficiency curves with respect to the power output are shown in Fig. 2.

The cost of fuel makes up a considerable amount of the generator's total cost.

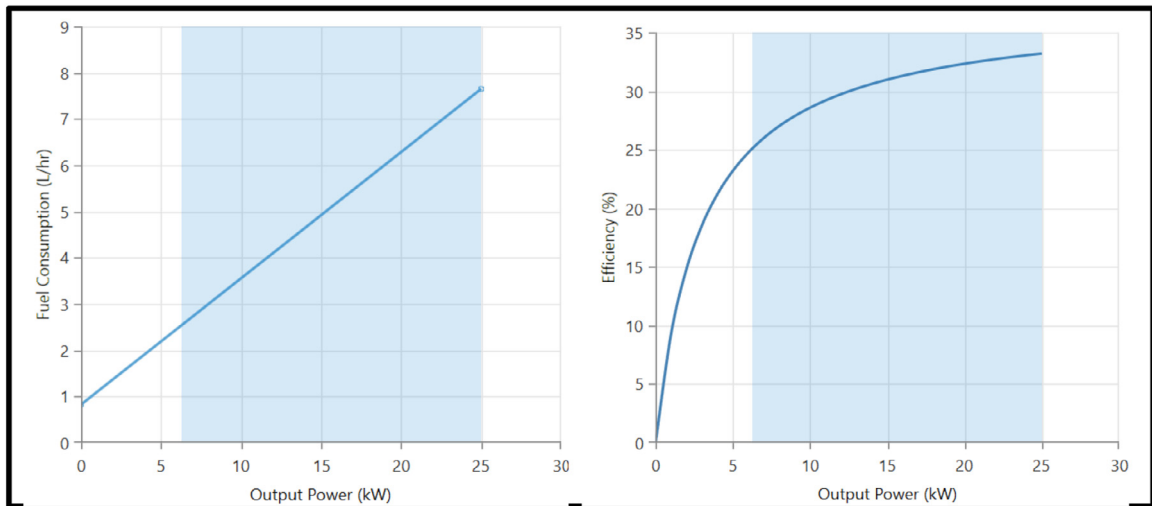


Fig. 2. Fuel consumption curve (left) and efficiency curve (right) of the DG.

Table 5

Summary of NPC, COE, Operation costs, and initial costs results for the optimal systems.

Configuration	NPC (\$)	COE (\$)	Operation costs (\$/yr)	Initial costs (\$)
PV /Battery	235,596	0.245	9424	82,013
Wind/Battery	1,429,946	1.49	31,860	910,694
PV/DG/Battery	233,693	0.238	9782	74,273
Wind/DG/Battery	404,976	0.414	20,285	74,373
PV /Wind/Battery	237,230	0.246	9236	86,698
PV/Wind/DG/Battery	236,456	0.24	9571	80,471

Battery system

Batteries are used to store excess solar and wind energy and to provide that energy when solar energy is unavailable. The batteries are also used to ensure that more renewable energy is incorporated into the load.

A 1KWh lead-acid battery with a 12 V operating voltage, a maximum capacity of 83.4 Ah, and an 80% roundtrip efficiency was used in the simulation. Each battery cost is set at \$300 for a 10-year life span, and the minimum state of charge is 100%, with a 40% initial state of charge.

Converter

In this study, a bi-directional converter is employed to convert DC power to AC power and vice versa. The PV and battery systems generate DC electricity, while the generator supplies AC power directly to the load. The ABB MGS100 60 kW converter was utilized. For simulation purposes, the inverter efficiency is taken to be 95% for a life period of 15 years, and the rectifier efficiency is assumed to be 95% for a relative capacity of 60%. The converter capital cost and replacement cost are considered as \$5000 and \$4800, respectively, and \$15 per year as O&M costs.

Optimization results and discussion

Despite various optimization results, the system with the lowest Net Present Cost (NPC) is chosen. Six hybrid solutions were considered in this analysis. After importing the data into the software and configuring the components, the optimization results are generated.

Economic analysis

The nominal discount rate for South Sudan considered in this study is considered as 15% adopted from [9] and the inflation rate of 11% was considered adopted from a forecast by O'Neill[10]. The diesel price is considered at \$0.28 per liter, a 2016 rate based on World Bank data from 2006 to 2016.

In comparison to other configurations, the PV/DG/Battery system is the most cost-effective optimal system, with an NPC of \$233,693 and a COE of 0.238 \$/kW, as shown in Table 5.

Table 6 summarizes the costs of the various configurations, with the optimal system's (PV/DG/Battery) capital costs, replacement costs, O&M expenses, fuel costs, and salvage costs being \$74,273, \$86,055, \$45,032, \$41,749.61, and \$13,416.72, respectively with a project lifespan of 25 years.

Table 6

Cost summary of various hybrid systems components.

Configuration	Component	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
PV	Converter	2813	1588	138	0	-371	4166
/Battery	Battery	36,000	97,435	39,115	0	-1921	170,629
	Solar System	43,200	0	17,602	0	0	60,802
Wind/Battery	Converter	14,594	8238	714	0	-1927	21,618
	Wind	708,000	308,078	38,463	0	-193,574	860,966
	Battery	188,100	187,233	204,374	0	-32,345	547,362
PV/DG/Battery	Converter	2928	1653	143	0	-387	4337
	Battery	30,600	77,877	33,247	0	-9737	131,987
	DG	12,500	6526	133	41,750	-3293	57,616
	Solar System	28,245	0	11,508	0	0	39,753
Wind/DG/Battery	Converter	673	380	33	0	-89	997
	Wind	54,000	23,497	2934	0	-14,764	65,667
	Battery	7200	23,218	7823	0	-805	37,436
	DG	12,500	57,431	777	230,470	-301	300,876
PV	Converter	2848	1607	139	0	-376	4218
/Wind/Battery	Wind	6000	2611	326	0	-1640	7296
	Battery	35,100	94,708	38,137	0	-2398	165,547
	Solar System	42,750	0	17,418	0	0	60,168
PV/Wind/DG/Battery	Converter	2946	1663	144	0	-389	4364
	Wind	6000	2611	326	0	-1640	7296
	Battery	30,600	77,683	33,247	0	-10,023	131,507
	DG	12,500	6169	123	38,342	-3852	53,282
	Converter	28,425	0	11,581	0	0	40,006

Table 7

Emissions from the hybrid combinations.

Configuration	Carbon Dioxide	Carbon Monoxide	Unburned Hydrocarbons	Particle Matter	Sulfur Dioxide	Nitrogen Oxides
Units	kg/yr	kg/yr	kg/yr	kg/yr	kg/yr	kg/yr
PV/Battery	0	0	0	0	0	0
Wind/Battery	0	0	0	0	0	0
PV/DG/Battery	8178	51	2.25	0.306	20	48
Wind/DG/Battery	45,146	282	12.4	1.69	111	265
PV /Wind/Battery	0	0	0	0	0	0
PV/Wind/DG/Battery	7511	46.9	2.07	0.281	18.4	44.1

Technical analysis

As shown in Table 7 the PV/wind/battery, PV/battery, and wind/battery combinations all have zero emissions because they all have a 100% renewable fraction. The Wind/DG/Battery system emits more emissions than the other systems in this study, owing to the low wind potential at the study site, which means that much of the system's demand energy must come from the DG.

The electrical production from the various configurations is summarized in Table 8. The annual power output from PV alone is sufficient to meet the annual AC primary demand, however, due to intermittency, there is a capacity shortage of 3.07%, with an unmet electrical load of roughly 2%. The optimal system has a renewable fraction of 85%, no unmet electrical load, no capacity shortage, and 47% surplus electricity. Wind/DG/Battery is the system with the least amount of excess electricity production. In general, PV systems generate a considerable portion of the electricity in any combination in which they are used, while DG comes in second in every combination with a PV system. When compared to PV and DG, wind output has the lowest output power.

The yearly total renewable energy output and the AC load demand are depicted in Fig. 3.

The generator's peak power output is generally between 18:00 and 23:00 h of the day, as seen in Fig. 4. The DG operation life is expected to be 18 years, with 817 h of operation per year.

As both the PV system and the battery system are connected to a DC bus, the selected battery system with a voltage of 12 V is connected in three string sizes to match the selected PV system voltage of 36. Fig. 5 depicts the state of charge and input power of the battery over a year.

Furthermore, the average state of charge of the battery storage system for the ideal combination as shown in Fig. 6 reveals the lowest charging cycles in July, followed by June, and then August. This is due to the low solar output during these months. The low solar output is attributed to the fact that June, July, and August are rainy season months with plenty of rainfall. As a result, cloud cover is prevalent during these months of the rainy season in the country.

Table 8
Production from the hybrid combinations.

Configuration	Production	PV System	Wind	DG	AC Primary Load	Excess Electricity	Unmet Electrical Load	Capacity Shortage	Renewable Fraction%	Max. Renewable Penetration%
PV/Battery	kWh/yr	178,838	0	0	59,072	111,703	1258	1855	100	4672
	%	100	0	0	100	62.5	2.09	3.07		
Wind/Battery	kWh/yr	0	185,516	0	58,914	120,417	1416	1855	100	19,788
	%	0	100	0	100	64.9	2.35	3.08		
PV/DG/Battery	kWh/yr	116,926	0	8974	60,330	59,149	0	0	85.1	3055
	%	92.9	0	7.13	100	47	0	0		
Wind/DG/Battery	kWh/yr	0	14,150	48,770	60,058	1191	272	839	0.188	1509
	%	0	22.5	77.5	100	1.89	0.451	1.39		
PV /Wind/Battery	kWh/yr	176,975	1572	0	59,090	111,636	1241	1856	100	4662
	%	99.1	0.881	0	100	62.5	2.06	3.08		
PV/Wind/DG/Battery	kWh/yr	117,672	1572	8237	60,330	60,807	0	2.88	86.3	3113
	%	92.3	1.23	6.46	100	47.7	0	0.0048		

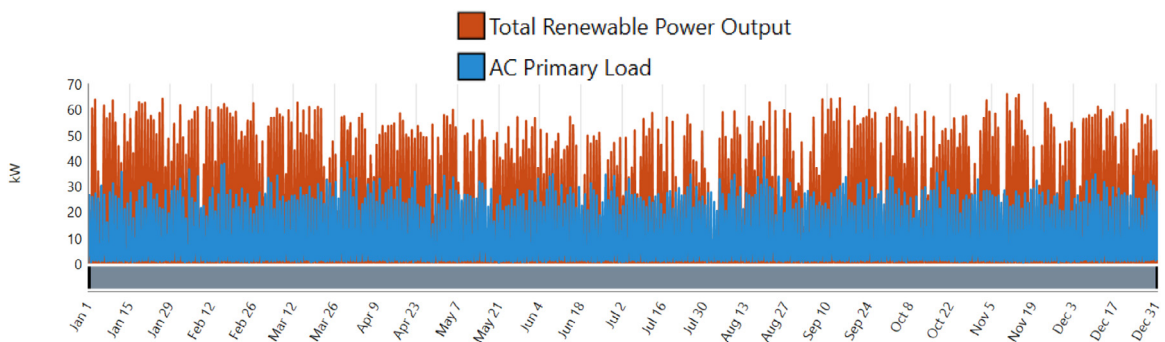


Fig. 3. Time Series plot for total renewable power output and AC primary load.

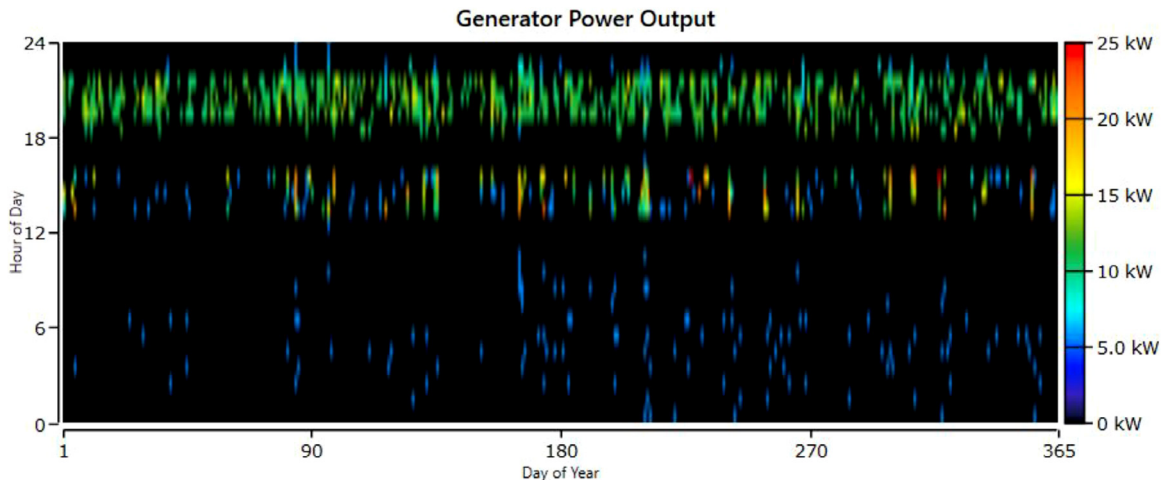


Fig. 4. Generator power output.

Conclusions

For rural electrification of isolated settlements in South Sudan's Southern region, the hybrid system is cost-effective. As per the findings of this study, the most cost-effective solution is a PV/DG/Battery arrangement. Excess electricity can be sold to the grid, sold to a neighboring community, or used as backup power in the event of a load demand spike. The optimized hybrid system can lessen the community's reliance on fossil fuels for energy generation, such as kerosene lights, while also

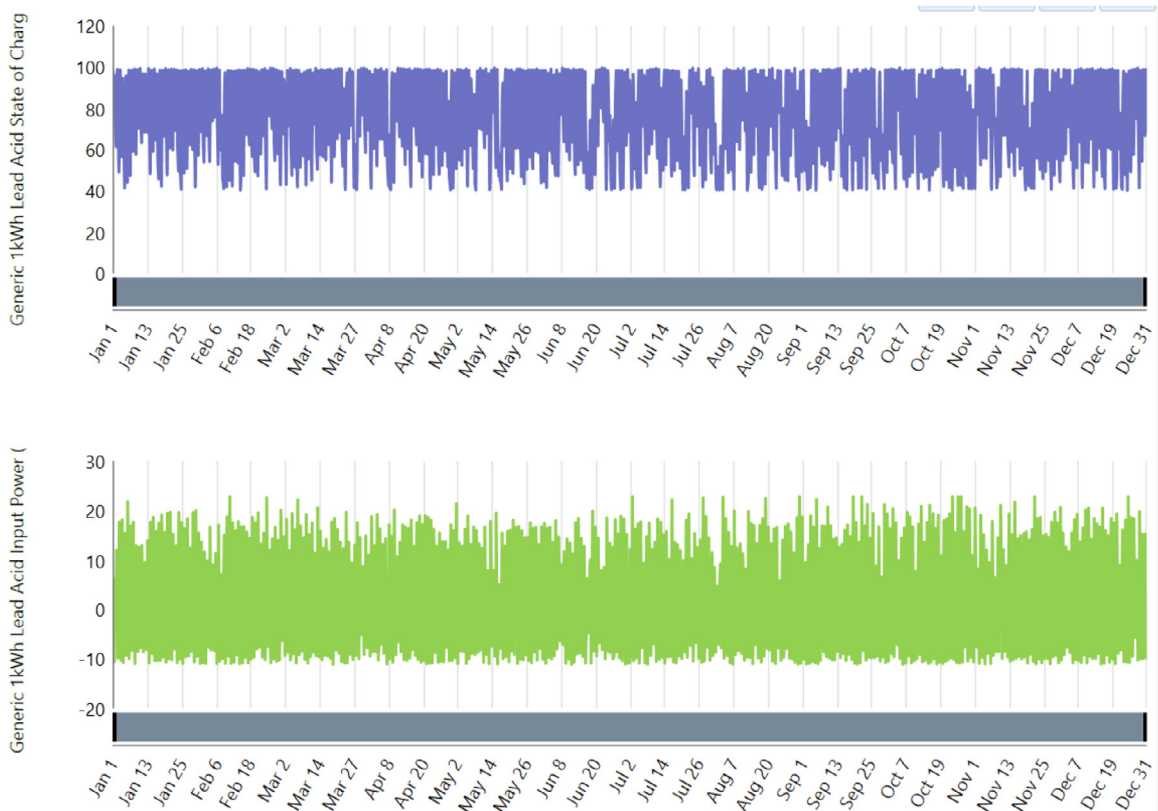


Fig. 5. Battery state of charge (indigo) and input power (green) for the optimal system.

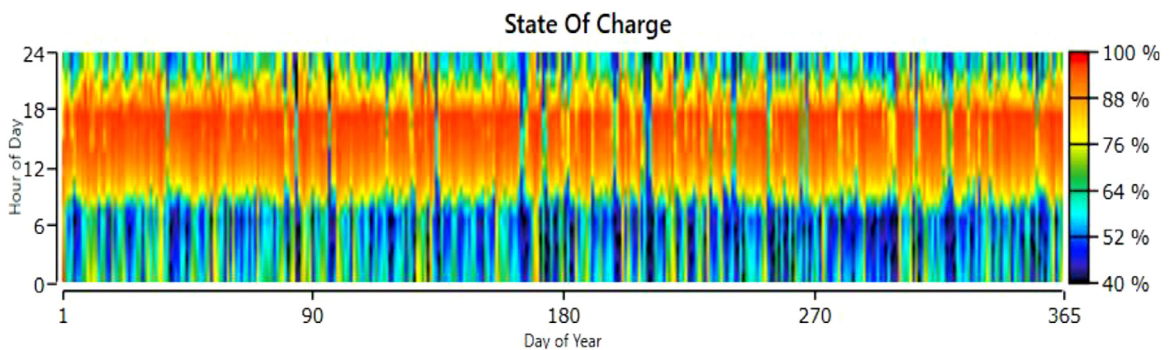


Fig. 6. The most feasible combination battery system state of charge.

preserving medical drugs and the optimized hybrid optimized system is capable of providing energy effectively to loads with enhanced reliability.

This study supports the South Sudanese government's goals to enhance electricity access for all by integrating renewable energy sources into the country's power system and reducing carbon emissions, which will benefit society, the economy, and the environment.

Furthermore, we anticipate load demand increase in the future as a result of various factors, including Payam's projected growing population and the acquisition of additional electrical appliances.

Finally, by disclosing the appropriate applicability and opportunity of renewable energy resources to the government and local or foreign entrepreneurs, this study is projected to add to the research of renewable energy potentiality and harnessing in South Sudan. It also provides a local-level analysis of hybrid power system feasibility in South Sudan for industrial operators and other stakeholders, highlighting which important drivers should be prioritized.

Data availability statement

Some or all data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request.

Item: the available data from the corresponding author is an excel model of the load profile (electrical appliances) of the case study.

Declaration of Competing Interest

None.

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